

Project Report: Customer Shopping Behaviour Analysis

1. Executive Summary

This project aims to analyze customer purchasing patterns and demographic trends using a dataset of 3,900 shopping records. By leveraging **Python** for data exploration, **SQL** for targeted business queries, and **Power BI** for interactive visualization, the analysis identifies key revenue drivers, customer segments, and product performance metrics to inform strategic business decisions.

2. Dataset Overview

The dataset (customer_shopping_behavior.csv) consists of **3,900 entries** and **18 features**, capturing various aspects of the customer journey:

- **Demographics:** Customer ID, Age, Gender, Location.
 - **Transaction Details:** Item Purchased, Category, Purchase Amount (USD), Payment Method.
 - **Customer Preferences:** Size, Color, Season, Review Rating.
 - **Behavioral Data:** Subscription Status, Shipping Type, Discount Applied, Previous Purchases, Purchase Frequency.
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3. Data Cleaning & Preprocessing

Significant steps were taken to ensure data quality and prepare it for analysis:

- **Missing Values:** Addressed minor gaps in Review Rating (found in 37 records).
 - **Feature Engineering:** - **Age Grouping:** Categorized customers into "Young Adult," "Middle-aged," and "Senior."
 - **Customer Segmentation:** Classified users based on purchase history:
 - **New:** 1 purchase.
 - **Returning:** 2–10 purchases.
 - **Loyal:** More than 10 purchases.
 - **SQL Integration:** Data was prepared and exported to a MySQL environment for advanced relational queries.
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4. Key Insights & Findings

4.1 Financial Performance

- **Total Revenue:** \$233,081
- **Average Purchase Value:** \$59.76

- **Revenue by Gender:** The customer base is predominantly **Male (68%)**, contributing significantly more to the total revenue compared to **Female (32%)** customers.
- **Top Categories by Revenue:**
 1. **Clothing:** \$104,264
 2. **Accessories:** \$74,200
 3. **Footwear:** \$36,093
 4. **Outerwear:** \$18,524

4.2 Customer Segmentation

The business boasts a high retention rate:

- **Loyal Customers:** 3,116 (79.9%)
- **Returning Customers:** 701 (18%)
- **New Customers:** 83 (2.1%)

Observation: A vast majority of the revenue comes from repeat, loyal customers, indicating high brand stickiness.

4.3 Product & Ratings

- **Top Selling Items:** Blouse, Jewelry, Pants, Shirt, and Dress are the most frequently purchased items.
- **Average Review Rating:** 3.75 / 5.0
- **Subscription Impact:** Approximately **27%** of customers are subscribed. SQL analysis suggests these customers tend to have more consistent purchasing patterns.

5. Technical Implementation

5.1 Python (Jupyter Notebook)

The Customer_Shopping_Behaviour_Analysis.ipynb file contains the Exploratory Data Analysis (EDA). It utilizes pandas for manipulation and matplotlib/seaborn for initial trend spotting. It also handled the conversion of the dataset into a format suitable for database insertion.

5.2 SQL Analysis

The customer_behavior.sql script addresses critical business questions, such as:

- Identifying customers who spend above average even when using discounts.
- Ranking top 3 products within each category using window functions (ROW_NUMBER).
- Comparing average spend between standard and express shipping to determine if faster shipping correlates with higher order values.

5.3 Power BI Dashboard

The Customer_Behaviour_dashboard.pbix provides a visual summary of these metrics, allowing stakeholders to filter data by Season, Category, and Location to see real-time shifts in consumer behavior.

6. Recommendations

1. **Targeted Marketing for Women:** Since the female segment is smaller (32%), there is a significant growth opportunity. Targeted campaigns or product lines tailored to female shoppers could balance the demographic split.
2. **Loyalty Programs:** While 80% of customers are loyal, the subscription rate is only 27%. Converting "Loyal" non-subscribers into "Subscribers" via exclusive perks could further stabilize revenue.
3. **Category Expansion:** Focus marketing efforts on "Accessories" and "Footwear," as they show strong performance but trail "Clothing" by a wide margin.
4. **Seasonal Optimization:** Use the "Season" data to manage inventory more effectively, ensuring top-rated items like "Blouses" and "Jewelry" are stocked according to seasonal demand peaks.